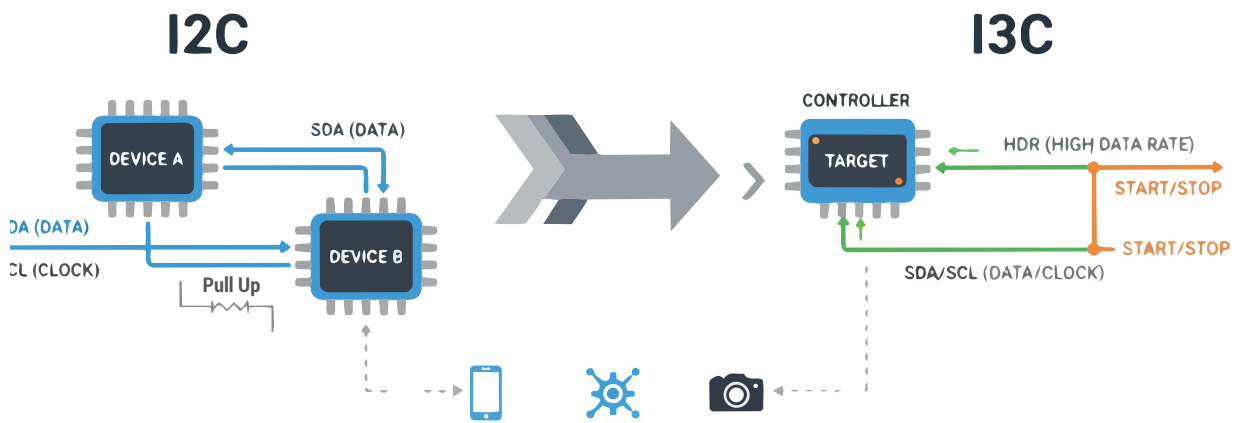


I2C and I3C: How Modern Devices Communicate

I3C and I2C are both two-wire communication protocols that help exchange data between multiple devices. While I3C preserves the simplicity of I2C, it introduces new features suited for today's sensor-rich devices.



Inside every electronic system, from smartphones and laptops to massive data centre servers, there are many small components or peripherals that must communicate with a central controller. These include sensors that monitor temperature and voltage, EEPROMs that store configuration data, and power management ICs.

All these peripherals work together to keep the system running smoothly. However, connecting each component with its own set of wires would make the board complex, messy and costly. To overcome this, engineers use shared communication buses, where devices use the same wires and take turns to communicate.

One of the simplest and most used protocols for this purpose is I2C (inter-integrated circuit) — a two-wire communication protocol to exchange data between multiple devices.

In server environments, I2C plays an essential role inside the baseboard

management controller (BMC), which is a dedicated chip responsible for monitoring the system health, reading sensors, managing power rails, and more. I2C acts as the communication path behind the BMCs to monitor and control different parts of a running server.

How I2C works

I2C is a bus interface connection protocol incorporated into devices for serial communication. It uses only two bi-directional open-drain lines (as shown in Figure 1):

- SDA (serial data line), which carries the actual data bits
- SCL (serial clock line), which

carries the clock signal

Each device connected to these lines acts as either a master (which initiates communication) or a slave/target (which responds).

A device can only pull the bus line to LOW (0), and it never drives the line HIGH (1). When the device releases the line, an external pull-up resistor pulls the line up to HIGH (as shown in Figure 2).

So, each device has two states for the bus pin:

- **Drive LOW:** Turn ON the transistor → line connects to GND → 0
- **Release:** Turn OFF the transistor → line floats → pull-up resistor makes it 1

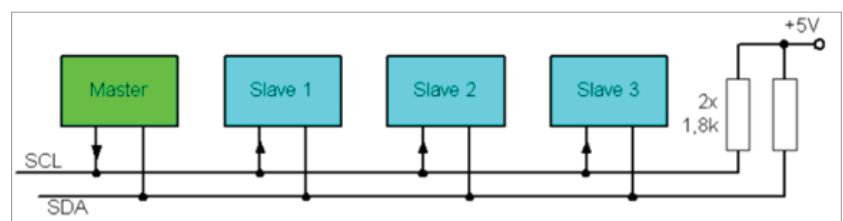


Figure 1: I2C connection